

CSE422 Assignment

(Submit within 19th September: <https://forms.gle/u5uQqH37uvKbzup8A>)

Question 1

Given two variables $X = \{\text{outcome of an unbiased dice that can be rolled to obtain an integer value between 1 and 6 with equal probability}\}$ and $Y = \{\text{outcome of an unbiased coin that can be tossed with an equal probability of heads and tails}\}$, Identify which is larger: entropy of X or entropy of Y ?

Question 2

In the table below, you are given a dataset containing 9 rows and 3 features X_1, X_2, X_3 . Y is the label. Using naive bayes classifier, Determine the most likely value of Y if $X_1 = 1, X_2 = a, X_3 = q$. You don't need to use any kind of smoothing or normal distribution. Just derive the probabilities from frequencies.

X_1	X_2	X_3	Y
1	a	p	0
2	b	r	1
3	b	p	1
3	c	q	1
2	c	r	0
1	b	q	1
2	a	p	0
3	a	r	1
3	b	q	0

Question 3

	A		A'	
	B	B'	B	B'
C	0.1	0.2	0.2	Y
C'	X	0.1	0.1	0.1

Using the given table answer the following questions:

- Assume the events A and B showcase absolute independence and $P(B|A) = 0.5$. Now determine the value of X
- Using the ans obtained from (I), determine the value of Y
- Using the ans obtained from (I) and (II), determine the value of $P(A|B \cap C)$

Question 4

Consider two medical tests, A and B , for a virus. Test A is 95% effective at recognizing the virus when it is present, but has a 10% false positive rate (indicating that the virus is present, when it is not). Test B is 90% effective at recognizing the virus, but has a 5% false positive rate. The two tests use independent methods of identifying the virus. The virus is carried by 1% of all people. Say that a person is tested for the virus using only one of the tests, and that test comes back

positive for carrying the virus. Which test returning positive is more indicative of someone really carrying the virus? Justify your answer mathematically.

Question 5

A study was conducted among 15 participants (10 Male, 5 Female) to assess the chances of being a smoker based on gender. It was seen that among males, the smoker to non-smoker ratio was 70:30, and it was 20:80 for females. Find Entropy(Smoking|Gender = male).

Question 6

ID	Rain	Temperature	Remaining topic	Class cancelled ?
1	yes	hot	A few	yes
2	yes	hot	A lot	yes
3	yes	normal	A few	yes
4	no	cold	A few	no
5	no	cold	A lot	no
6	yes	cold	A few	yes
7	yes	cold	A lot	no

- Find the root node of a decision tree on “Class cancelled ?” from the table above

-If ‘Temperature’ feature is in the root node, in its ‘cold’ branch, what will be the next prioritized feature according to the ID3 algorithm?

Question 7

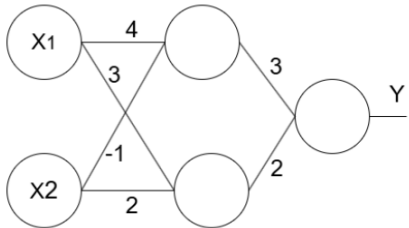
X	Y
2	11
4	23
6	28

Suppose, given a set of data samples in the table above, you are to fit the data points with a straight-line hypothesis $Y = mX + c$, where m is the slope and c is the y-intercept. The values of X and Y come from the data samples.

-If the initial value of m is 9 and c is 2, calculate the error produced by the proposed hypothesis mean square error (MSE) or sum of square residual (SSR) function.

-Update the m and c values with one iteration of gradient descent. If applicable, you can assume a learning rate of 0.001

Question 8



-If $X_1 = 5$ and $X_2 = 1$, apply forward propagation to calculate the value of Y . Consider that the hidden layer uses ReLU activation function while the output layer uses the sigmoid activation function.

-Mention the importance of activation functions in a neural network

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Question 9

-Describe the differences between regression and classification.

-What should you do regarding the learning rate if you are close to the global minima while your model performance is fluctuating up and down?

-In the mean square error loss function, why do we omit the negatives by squaring it, instead of using absolute?

-Why entropy is a better loss function than mean square error for classification problems?

Question 10

X	Y
0.2	0
0.4	0
0.8	1

-If you fit the above data samples using $Y=mx+b$, where $m = 3$ and $b = 1$, calculate the amount of error using binary crossentropy loss function.

-Perform one iteration of gradient descent to update both m and b , assuming that the learning rate is 0.1